

Frequency distribution tables

By size and month

Task 3: Estimate the 90% confidence interval for the same task as the lesson. What changes can you see?

Solution:

Women shoe sales

US	Germany, GER1												Germany, GER2												Mean	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11	12	GER1	GER2
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00	0.00
4.5	0	0	0	0	1	3	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0.42	0.08
5	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0.17	0.17
5.5	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	2	0	1	0	0.08	0.33
6	0	2	0	0	0	0	0	0	0	0	0	0	0	1	3	1	2	0	0	0	0	0	0	0	0.17	0.58
6.5	3	3	1	2	1	0	2	0	2	1	3	4	2	0	2	1	2	0	1	2	1	3	0	0	1.83	1.25
7	0	3	3	4	1	0	1	0	2	0	0	1	0	0	4	1	3	1	1	1	3	1	4	0	1.25	1.58
7.5	1	2	4	1	2	6	4	3	5	8	2	1	2	1	1	3	2	7	9	8	14	8	6	3	3.25	5.33
8	6	10	3	9	1	3	6	8	3	12	3	9	13	6	5	13	5	3	11	6	6	9	8	3	6.08	7.33
8.5	10	10	10	7	14	4	7	7	4	8	7	9	8	5	10	4	5	5	9	7	3	7	9	8	8.08	6.67
9	1	3	8	6	3	1	4	4	0	2	4	2	5	2	2	9	3	1	1	7	2	1	4	2	3.17	3.25
9.5	4	1	2	1	2	2	2	4	5	2	3	2	0	1	1	0	1	2	2	1	7	2	4	2	2.50	1.92
10	0	1	1	1	1	1	3	1	0	0	0	1	0	1	1	0	0	0	2	3	0	2	0	0	0.83	0.75
10.5	1	0	0	0	2	2	4	1	0	3	1	1	0	2	0	0	0	1	0	0	0	0	2	1	1.25	0.50
11.5	0	0	0	2	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	1	0	0	0	0	0.17	0.50
Total	26	35	32	33	28	22	35	28	21	36	25	30	30	19	30	35	20	24	35	38	35	36	37	24		

Sample variance		Pooled variance	Margin of error	90% CI	
GER1	GER2				
0.00	0.00	0.00	0.00	0.00	0.00
0.81	0.08	0.45	0.47	-0.13	0.80
0.33	0.33	0.33	0.40	-0.40	0.40
0.08	0.42	0.25	0.35	-0.60	0.10
0.33	0.99	0.66	0.57	-0.99	0.15
1.61	0.93	1.27	0.79	-0.20	1.37
2.02	2.27	2.14	1.02	-1.36	0.69
4.93	16.06	10.50	2.26	-4.35	0.18
12.27	12.24	12.25	2.44	-3.69	1.19
7.72	4.97	6.34	1.76	-0.34	3.18
5.06	6.57	5.81	1.68	-1.77	1.60
1.55	3.72	2.63	1.13	-0.55	1.72
0.70	1.11	0.91	0.66	-0.58	0.75
1.66	0.64	1.15	0.75	0.00	1.50
0.33	2.09	1.21	0.77	-1.10	0.44

	GER1	GER2
n	12	12
$t_{90\%,22}$		1.71

For a single shoe size (marked), we can clearly see that one shop outperforms the other.
However, our confidence is lower.
This shows you that the level of confidence does make a difference.
Also notice that the intervals are narrower.
Overall the two shops are extremely similar in terms of sales.